

Assignment – 7

Q1. Write a Java program to establish a connection with an RDBMS.

```
package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.util.TimeZone;

class Main {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";
            Connection connection = DriverManager.getConnection(url, username, password);

            if (connection != null) {
                System.out.println("Connected...");
                connection.close();
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

Connected...

Q2. Write a Java program to create a table named student in the connected database with the following fields: roll_no (INT, Primary Key), name (VARCHAR(50)), age (INT), department (VARCHAR(30)), cgpa (FLOAT(4,2))

```
package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.TimeZone;

class Main {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
```

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```

String username = "dream";
String password = "dream2503";
Connection connection = DriverManager.getConnection(url, username, password);

String query = ""
    CREATE TABLE student (
        roll_no INT PRIMARY KEY,
        name VARCHAR(50),
        age INT,
        department VARCHAR(30),
        cgpa NUMERIC(4, 2)
    );"";

Statement statement = connection.createStatement();
statement.executeUpdate(query);
System.out.println("Table 'student' created successfully!");

statement.close();
connection.close();
} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

Output:

Table 'student' created successfully!

Q3. Write a JDBC program to insert two student records into the student table with values given inside the program (hard-coded).

```

package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.TimeZone;

class Main {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";
            Connection connection = DriverManager.getConnection(url, username, password);

            String query = "INSERT INTO student VALUES (?, ?, ?, ?, ?)";
            PreparedStatement ps = connection.prepareStatement(query);

```

```

        ps.setInt(1, 101);
        ps.setString(2, "Rahul");
        ps.setInt(3, 20);
        ps.setString(4, "CSE");
        ps.setDouble(5, 8.5);
        ps.executeUpdate();

        ps.setInt(1, 102);
        ps.setString(2, "Anita");
        ps.setInt(3, 21);
        ps.setString(4, "ECE");
        ps.setDouble(5, 9.1);
        ps.executeUpdate();

        System.out.println("Two student records inserted successfully!");
        ps.close();
        connection.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Output:

Two student records inserted successfully!

Q4. Write a JDBC program to insert a student record into the student table. Take inputs for roll_no, name, age, department, and cgpa from the key board using Scanner.

```

package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;
import java.util.TimeZone;

public class Update {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";
            Connection connection = DriverManager.getConnection(url, username, password);

            Scanner sc = new Scanner(System.in);
            String query = "INSERT INTO student (roll_no, name, age, department, cgpa) VALUES
            (?, ?, ?, ?, ?)";

```

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```

PreparedStatement ps = connection.prepareStatement(query);
System.out.print("Enter your Roll no.: ");
ps.setInt(1, sc.nextInt());
System.out.print("Enter your Name: ");
ps.setString(2, sc.next());
System.out.print("Enter your Age: ");
ps.setInt(3, sc.nextInt());
System.out.print("Enter your Branch: ");
ps.setString(4, sc.next());
System.out.print("Enter your CGPA: ");
ps.setDouble(5, sc.nextDouble());
int rows = ps.executeUpdate();

if (rows > 0) {
    System.out.println("Record inserted successfully!");
}
ps.close();
connection.close();
} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

Output:

```

Enter your Roll no.: 11
Enter your Name: Swapnaraj
Enter your Age: 21
Enter your Branch: CSE
Enter your CGPA: 9.98
Record inserted successfully!

```

Q5. Write a JDBC program to display all records from the student table.

```

package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.TimeZone;

public class Select {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";

```

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```

        Connection connection = DriverManager.getConnection(url, username, password);

        String query = "SELECT * FROM student";
        Statement statement = connection.createStatement();
        ResultSet resultSet = statement.executeQuery(query);
        System.out.println("Student Records:\n");

        while (resultSet.next()) {
            System.out.println("Roll No: " + resultSet.getInt("roll_no"));
            System.out.println("Name: " + resultSet.getString("name"));
            System.out.println("Age: " + resultSet.getInt("age"));
            System.out.println("Department: " + resultSet.getString("department"));
            System.out.println("CGPA: " + resultSet.getFloat("cgpa"));
            System.out.println("-----");
        }
        resultSet.close();
        statement.close();
        connection.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Output:

Student Records:

Roll No: 101
 Name: Rahul
 Age: 20
 Department: CSE
 CGPA: 8.5

Roll No: 102
 Name: Anita
 Age: 21
 Department: ECE
 CGPA: 9.1

Roll No: 11
 Name: Swapnaraj
 Age: 21
 Department: CSE
 CGPA: 9.98

Q6. Write a JDBC program to update the CGPA of a student based on their roll_no. Take the targeted student's roll number and updated cgpa as input from the keyboard.

```
package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;
import java.util.TimeZone;

class Main {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";
            Connection connection = DriverManager.getConnection(url, username, password);

            Scanner sc = new Scanner(System.in);
            String query = "UPDATE student SET cgpa = ? WHERE roll_no = ?";
            PreparedStatement ps = connection.prepareStatement(query);

            System.out.print("Enter Roll No to update: ");
            ps.setInt(2, sc.nextInt());
            System.out.print("Enter new CGPA: ");
            ps.setDouble(1, sc.nextDouble());
            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("CGPA updated successfully!");
            } else {
                System.out.println("No student found with given roll number.");
            }
            ps.close();
            connection.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

```
Enter Roll No to update: 11
Enter new CGPA: 10
CGPA updated successfully!
```

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Q7. Write a JDBC program to delete a student record based on roll_no. Accept the roll number from the keyboard.

```
package assignment7;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;
import java.util.TimeZone;

class question7 {
    public static void main(String[] args) {
        try {
            Class.forName("org.postgresql.Driver");
            TimeZone.setDefault(TimeZone.getTimeZone("Asia/Kolkata"));
            String url = "jdbc:postgresql://localhost:5432/database";
            String username = "dream";
            String password = "dream2503";
            Connection connection = DriverManager.getConnection(url, username, password);

            Scanner sc = new Scanner(System.in);
            String query = "DELETE FROM student WHERE roll_no = ?";
            PreparedStatement ps = connection.prepareStatement(query);
            System.out.print("Enter Roll No to delete: ");
            ps.setInt(1, sc.nextInt());
            int rows = ps.executeUpdate();

            if (rows > 0) {
                System.out.println("Record deleted successfully!");
            } else {
                System.out.println("No student found with given roll number.");
            }
            ps.close();
            connection.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output:

Enter Roll No to delete: 11
Record deleted successfully!

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